Design a Microblogging System (Threads)

“Care about what other people think and you will always be their prisoner.”
— Lao Tzu

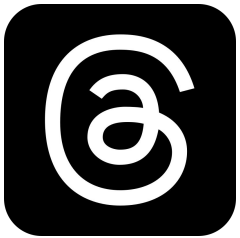


Outline

1. Understand The Problem
2. Estimation
3. High Level Design
4. Low Level Design

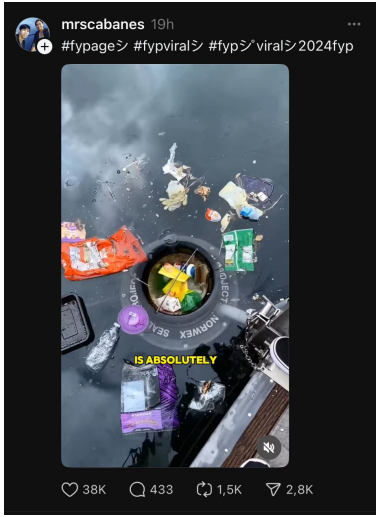
1. Understand The Problem

Ask Me!



1.1. Functional Requirements

- User B can follow user A
- User A can publish a post
  - The feed contains text only
- User B receive a notification about the post
- Build a feed for every user
  - Keep it simple, the feed is sorted by reverse chronological
- User B can react to the post
  - Like, comment, share, …
  - Show the number of reacts



1.2. Capacity

Questions:

1. How many users? How many DAU?
2. How many post does a user publish per day?
3. What is the average number of comments per post?
4. **Which posts the most users care about?**

Answers:

1. 500M users, 50M DAU
2. 1 post/DAU/day
3. 10 comments/post
4. Posts in the last 7 days OR trendy posts

1.3. Non-functional Requirements

- **Scalability**
- **Performance**
  - **Latency:**
    - Return a feed quickly (< 100ms). Building feed is aggregating so much data (posts, users, counters, …)
    - The post needs to be appear in followers' feed in short period of time.
  - **Throughput**
    - Normal: 1 QPS, Peak: 5M QPS
- Availability
- Security
- Eventually Consistency

2. Estimate

2. Estimate

- Assume that:
  - 3 KB/post
  - 500 B/post
- Storage for posts and comments in 5 years:

(3+0.5*10)*5 * 10^(3+7) * 400 * 5

= 8*5*4*5 * 10(3+7+2)

= 4*2 * 10^(3+7+2+2)

= 800 TB

- Cache
  - TTL: 7 day
  - Size: 3*5*7 * 10^(3+7) = 105 * 10^10 = 1.05 TB

- 500M users, 50M DAU
- 1 post/DAU/day
- 10 comments/post
- Posts in the last 7 days & trendy posts

3. High Level Design

Data Modeling

- Table: Post
  - Post_id (pk)
  - creator_id
  - content
  - status
  - created_at
  - updated_at
- Partition key: post_id

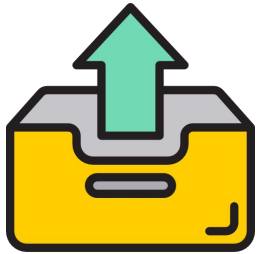
- Table: Feed
  - User_id (pk)
  - Post_id (pk)
  - status
  - order
  - created_at
- Partition key: user_id

Let's Design by Yourself

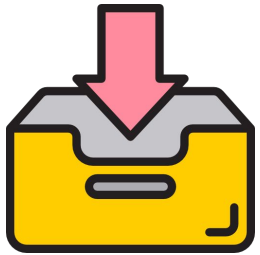
(mini interview)

Push Model

- Publisher **push actively the post to her followers**.
- Each user has **an inbox and outbox**. The system is like a postman.
- When a user creates a post - placing a letter into outbox, then a postman (the system) delivers to the inbox of receiver (follower).
- The most of a feed is built based on the inbox.



Outbox Publisher



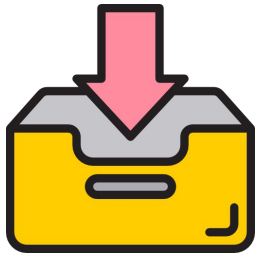
Inbox Follower

```
insert into outbox(userId, postId, create_time) values("A", $feedId, $current_time);  // write to A outbox
insert into inbox(userId, postId, create_time) values("B", $feedId, $current_time);  // write to B inbox
insert into inbox(userId, postId, create_time) values("C", $feedId, $current_time);  // write to C inbox
insert into inbox(userId, postId, create_time) values("D", $feedId, $current_time);  // write to D inbox
```

Push Model / Pros and Cons

- Pros:
  - Simple
  - **Building a feed is fast**
- Cons:
  - If a celebrity (a famous singer) has **a lot of fans**, then the process of sending to all inboxes is **very long**.
  - **Redundant push** (computing, resource) because of inactive users.
  - **Weak scalability**:
    - Unfollowing → deleting posts in inboxes
    - Users can follow categories: travel, tech, …



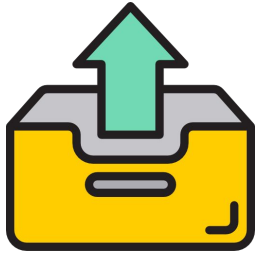


Inbox
Follower

Pull Model

- When building a feed, followers active pull posts from the outbox of publisher
- Pros:
  - Do not take time to broadcast.
  - **No redundant storage** because only actives users build feeds
  - **Scalability**
- Cons:
  - **Take time to build a feed** due to another calls.
  - **Complex** due to aggregate.





Outbox Publisher

So which one do you choose?
Push or Pull?

I prefer Pull model.
But how about latency?
I will try to handle it.
But how?

4. Low Level Design

Mixing Model: Push + Pull

- How to handle with latency?
  - If publisher is a normal user → Push
  - If publisher is a celebrity → Pull
  - Cache for latest posts
- How to limit redundant storage, computing resource?
  - Tracking Active Users

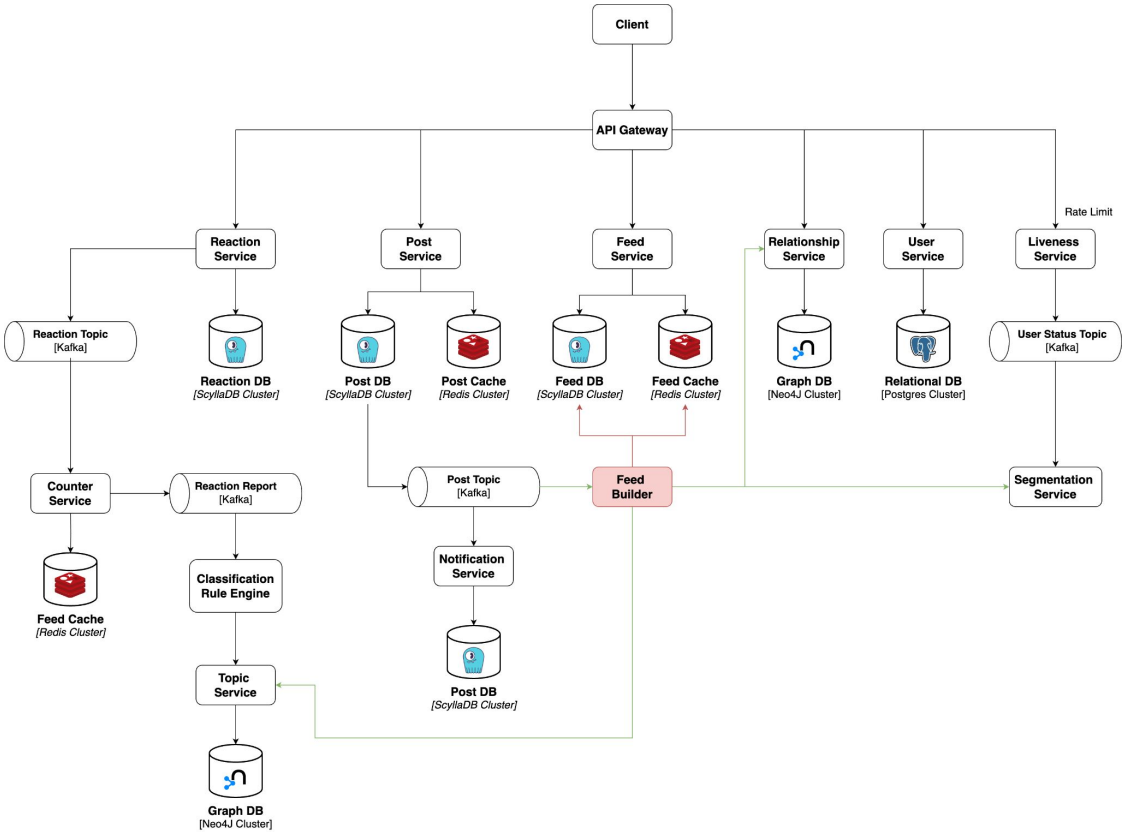
How to Track Active Users?

- Definition of “active users”? Depends on biz logic
- Sending heartbeats → Liveness Component
- What is heartbeat?
  - Opening app
- Problem: too many hits on system
  - Solution: Rate limit.
    Cache <user_id, last_active_ts>
    TTL: 1 day

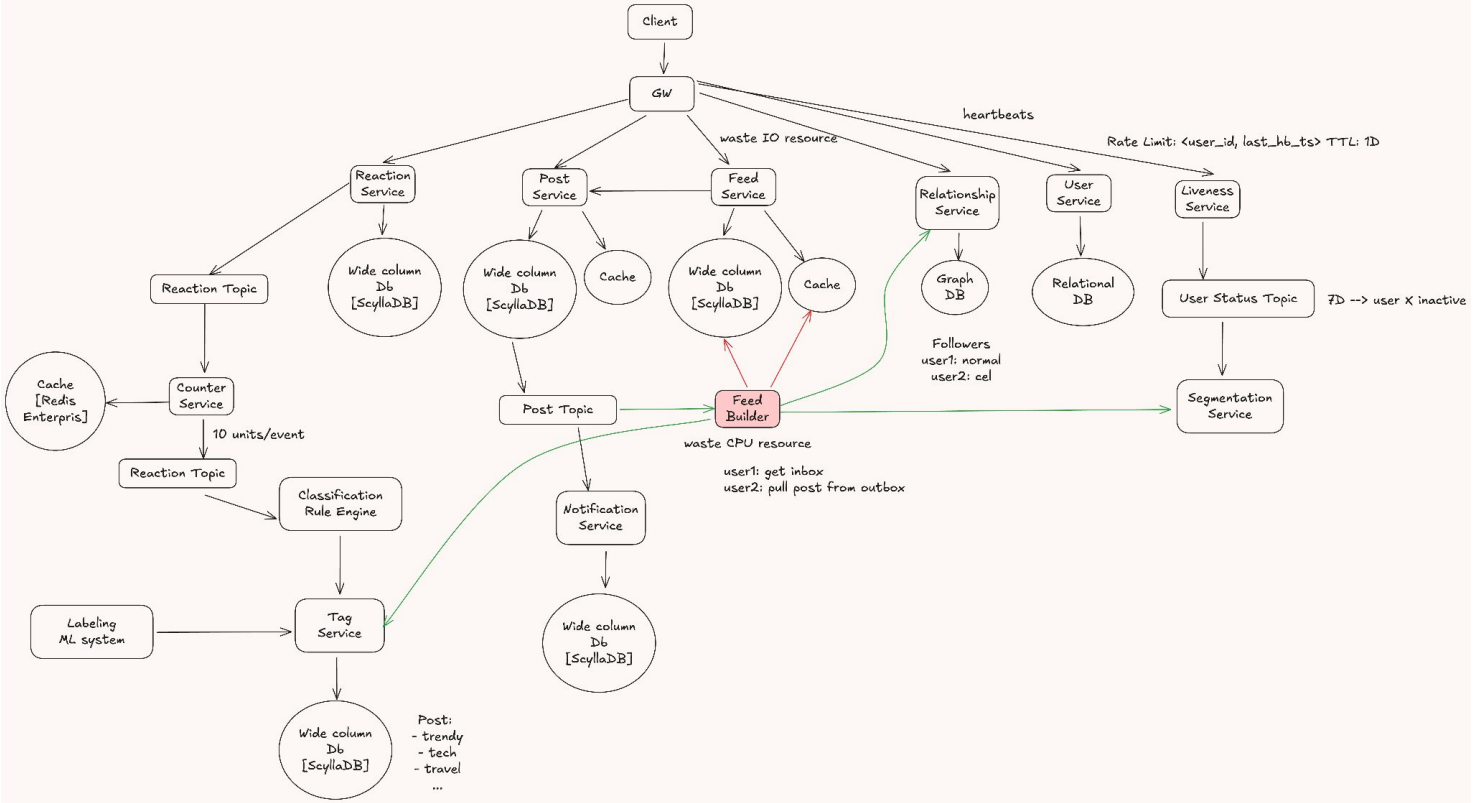
Trendy Post

- Definition of “Trendy”? Depends on biz logic
- Based on metrics: likes, comments, shares, …
- We also have other classification:
  - Tech
  - Business
  - Health
  - …

Final Design



Final Design

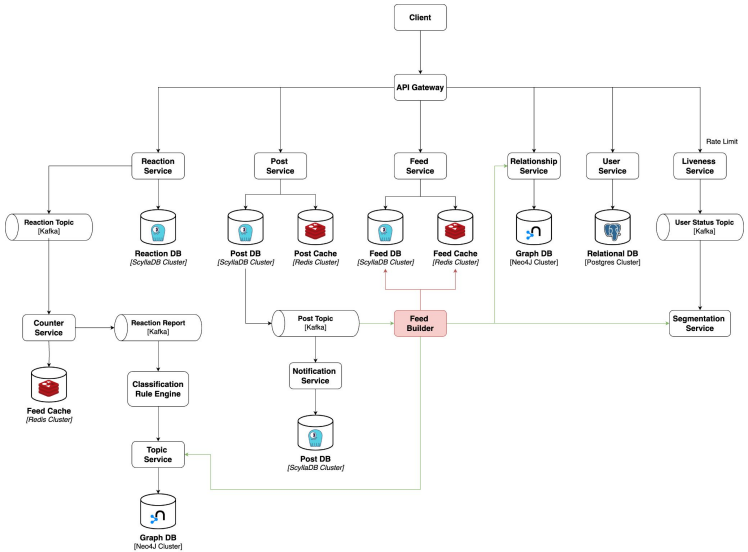


Recap

- User experience, user behavior affect to the design.
- Trade off: Push vs Pull.

  In this case, pull model is preferred due to scalability.
- Tracking active users is used to save storage space,

  but it make system more complicated.



Homework

- Design a Messaging System (Discord)

References

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Thank you 🙏

